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People of the sun: Local resistance and solar energy (in)justice in southern Portugal

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ABSTRACT

This paper explores a public controversy surrounding a large solar power plant planned for the village of Cercal, in Alentejo, southern Portugal, mobilizing energy justice literature to shed light on the contradictions and injustices (re)produced by hegemonic energy transitions. Drawing on qualitative data from interviews, fieldwork and document analysis, this controversy is unpacked through three main tenets of energy justice: distributional, procedural and recognition, contributing to current research on solar energy justice.

Our empirical findings indicate that the local population and environmental movements are concerned that Cercal might become a "green sacrifice zone" for the sake of the hegemonic energy transition. Procedural injustices were of particular concern, with interviewees claiming that the public consultation process failed to involve the local community, indicating that the State and corporate actors do not recognize residents as legitimate stakeholders, disregarding protest and active discord as legitimate forms of political participation.

This original case study, unfolding in a country currently experiencing a significant lithium mining controversy, is a flagship example of how top-down and corporate energy transitions are major drivers of energy injustices, enforcing green grabbing and the enactment of green sacrifice zones under the aegis of carbon reductionism. In order to address these contradictions, a radical shift in the socioeconomic and political dynamics underlying the hegemonic energy transition is urgently needed, alongside the meaningful involvement of local communities in decision-making processes directly affecting their territories and livelihoods.

1. Introduction

Drawing on a theoretical framework anchored in energy justice (EJ) literature, this paper explores a controversy surrounding the installation of a solar power plant (SPP) in the village of Cercal, Alentejo, a region in southern Portugal. Through this case study, we aim to illustrate the contradictions of the Portuguese energy transition. We reveal its failure to ensure inclusive and representative procedures in the deployment of renewable energy (RE) projects and to recognize local communities' concerns, aspirations and livelihoods, leading to an unequal distribution of burdens and benefits between the local community and the private company in charge of the project.

Burgeoning research indicates that global energy transitions, including at the European level, are driving green extractivism and green grabbing [1–9]. The energy transition is intensifying "market relationships, extraction, and infrastructural colonization" [10], ultimately strengthening global capitalism, itself at the roots of the climate crisis. Such extractive practices are often met with local resistance, including in Portugal, which is currently struggling with a controversy around lithium mining [11,12] - indeed, the Portuguese environmental movement has strongly reacted against this so called "green mining". RE projects, particularly large solar power plants, such as the one in Cercal, are also being contested [13], as they are driving green grabbing, "the appropriation of land and resources for environmental ends" [14,p.237].

Regarding the Portuguese government's goals for the energy transition, the first version of the revised National Energy and Climate Plan (PNEC - Plano Nacional de Energia e Clima) 2030 [15], released in June 2023, sets a target of 85 % of renewable electricity by 2030, while in 2021 it was situated around 60 % (Please see Image 1). The most significant increase is expected to come from photovoltaic solar energy:

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while Portugal currently has 2.6 GW in operation, by 2030 it should have 20.4 GW. In the revised PNEC, the government estimates 14.9 GW of centralized photovoltaic production (more than double the 7 GW of the initial PNEC, and much higher than the 1.5 GW currently in operation in large solar plants) [15].

Considering these goals, the Cercal project is one of many large-scale solar power plants planned to be implemented over the next decade. Therefore, it is important to assess what achieving the targets set by the Portuguese government entails at the local scale. More specifically, in a context of expected acceleration in the deployment of large-scale solar power plants in the Alentejo region, it is essential to explore their impacts and the potential injustices that may result from their implementation.

Based on a qualitative approach, supported by document analysis, fieldwork and semi-structured interviews, we attempt to answer the following research question: does the deployment of the Cercal SPP meet the three tenets of energy justice? Three sub-questions derive from the main research question: 1) how does this case study illustrate the contradictions of the Portuguese energy transition? 2) How does this controversy relate to recent concerns surrounding the creation of green sacrifice zones in Portugal? 3) What are the limits of institutional processes of consultation and public participation?

We argue that the RE project planned for Cercal fails to address three relevant tenets of EJ – distributional, procedural and recognition justice –, illustrating how certain regions in Portugal and in Europe's semi-periphery [9] are becoming “green sacrifice zones” [16] in the name of climate action and under the aegis of carbon reductionism [17]. Green Sacrifice Zones refers to the fact that “the logic of sacrificing a certain space or ecology can be expanded to include places and populations that will be affected by the sourcing, transportation, installation, and operation of solutions for powering low-carbon transitions, as well as end-of-life treatment of related material waste” [16,p.543]. These, in turn, are related to carbon reductionism, the bottlenecking of socioenvironmental issues (for instance, local ways of life and livelihoods, or biodiversity conservation) into the single goal of carbon dioxide emissions reduction [17]. Furthermore, we argue that formal public consultation procedures are ineffective participation mechanisms, as they are not binding and do not provide citizens with any decision-making power.

The academic and sociopolitical contribution of this article is threefold: 1) it adds to a growing body of literature on solar EJ, by examining an original case study; 2) it is the first empirical study applying the EJ framework to a RE project in Portugal; 3) since the case study is an ongoing controversy, this article may also contribute to the public debate surrounding RE deployment, and the structural limitations of Portuguese formal public participation processes.

The article is structured as follows: in the following section, we review the literature on energy justice, specifying how this theoretical framework will be applied in this paper; in the methodology section, the case study is characterized and the process of data collection and analysis is detailed; subsequently, the data is systematized according to three tenets of energy justice; the results are then discussed in light of relevant literature; in the conclusion, we advance avenues for future research, as well as a set of policy recommendations.

2. Energy justice

EJ scholarship emerged at the intersection of socioecological justice and energy systems. In line with Sovacool et al. [18,p.5], we acknowledge that energy system interventions, namely the deployment of RE projects, entail more than just technology and economic development – they are underpinned by sociopolitical structures and power relations, raising moral and ethical concerns regarding equity and justice. However, as contended by Jenkins et al. [19,p. 67], considerations of equity and justice are underrepresented within the energy transitions debate. To fill this gap, EJ has been incrementally considered an important analytical and decision-making tool for both researchers and energy planners and consumers [20,p.436].

The first meaningful attempt to systematize literature on justice and energy was offered by McCauley et al. [21,p.108], who argue that energy policy needs to address “the unequal distribution of ills” from decisions on infrastructure siting, subsidies, pricing, and consumption indicators within the context of global and local pressures. Influenced by environmental justice literature, which, in turn, builds on Nancy Fraser's theory of social justice [22,23], McCauley et al. [21,p.108–109]

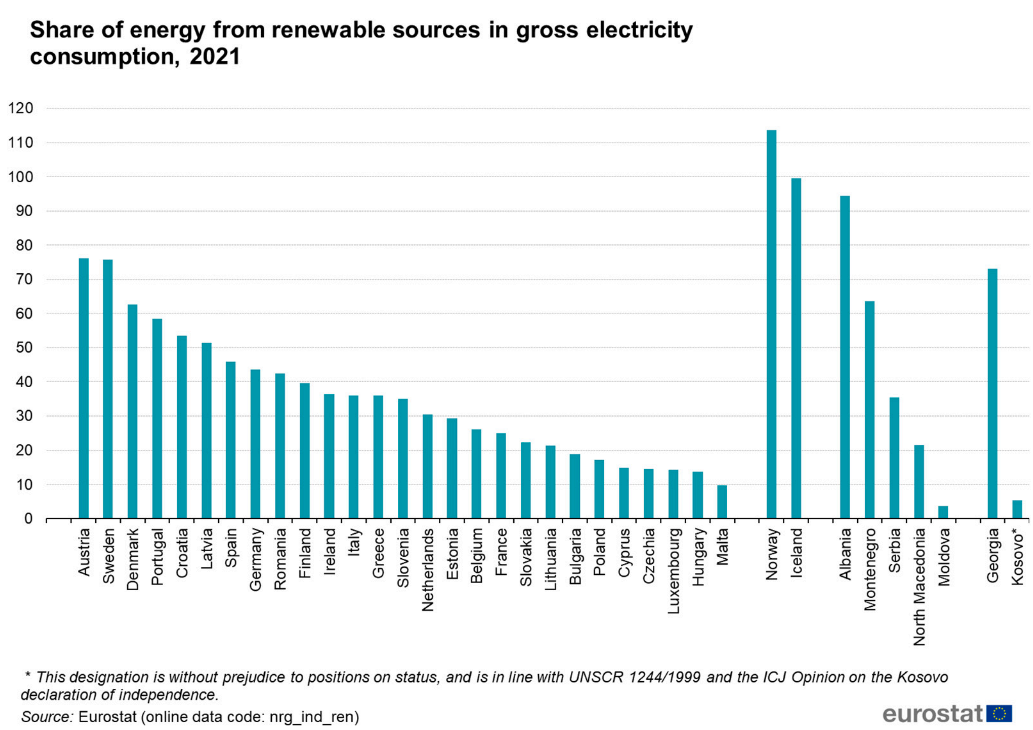

Image 1. European countries' share of energy from renewable sources in gross electricity consumption in 2021. Source: Eurostat.

advanced three main tenets of EJ: 1) Distributional justice, which describes "the physically unequal allocation of environmental benefits and ills and the uneven distribution of their associated responsibilities", thus calling for an equal distribution of benefits and ills among all members of society; 2) Procedural justice, which refers to equitable and non-discriminatory procedures that engage all stakeholders, requiring broad, inclusive, and effective participation in decision-making, transparency from government and industry, and suitable engagement mechanisms; 3) Recognition justice, which states that individuals must be "fairly represented", "free from physical threats", and in possession of full and equal political rights. Lack of recognition can materialize in "cultural and political domination, insults, degradation and devaluation", as well as misrecognition, more specifically, "a distortion of people's views that may appear demeaning or contemptible" [21,p. 109].

Thus, EJ evaluates (1) where injustices emerge in the world, (2) which affected social groups are ignored or misrepresented, (3) which processes exist for their remediation to reveal and reduce such injustices [24]. A new transversal pillar was eventually added to the "triumvirate of tenets" – restorative justice [25,26]. According to Hazrati and Heffron [26,p.1], the incorporation of restorative justice in energy justice would have a preventive effect, as decision making would, by design, internalize any potential injustices, harms or damages that could result from it.

To enhance the conceptual and analytical potential of EJ, Sovacool and Dworkin [20] developed an EJ decision-making framework rooted in eight central principles: availability, affordability, due process, good governance, sustainability, intergenerational equity, and responsibility. Later, Sovacool et al. [27,p.687] added resistance and intersectionality, thus recognizing how EJ is intertwined with race, class, power and the treatment of non-humans.

Nevertheless, several authors challenge singular understandings of EJ. LaBelle [28,p.615], for instance, contrasts two definitions: universal EJ and particular EJ. By the same token, Mejía-Montero et al. [29,p.10] warn against universalized understandings and constructions of EJ, favoring contextualized ones, given that energy systems are embedded in preexisting social, environmental, economic, and cultural contexts. Similarly, Tornel [30,p. 44] calls for a decolonization of EJ, as it "tends to reproduce rather than transform hegemonic power relations". Such critiques do not weaken EJ's conceptualization – Iwinska et al. [31,p.11] claim that the flexibility and contextual adaptability of EJ seems to be its strength, while also ensuring the stability and coherence of its core definition. Energy justices, in the plural form, acknowledge the political, social, economic, environmental, and cultural embeddedness of energy systems, their distinct features, and the potential injustices arising from energy transitions. Thus, Sovacool et al. [27] underline the urgency of resisting unjust energy projects and policies and understanding the complex intersectionality of justice concerns, acknowledging that EJ cannot be detached from biodiversity loss, as well as socioeconomic issues like globalization, economic inequality, immigration, health, and security.

2.1. Energy justice in renewable energy transitions

Even though the transition to RE systems is a key condition for ending fossil fuel dependency, drastically cutting greenhouse gas emissions and mitigating the climate crisis, it is not exempt from multiscalar and multidimensional injustices, especially if the underlying sociopolitical relations remain asymmetrical. We contend that the deployment of RE projects is one of the settings where these injustices can emerge, since they can result in conflicting interests between the actors involved (for instance, national governments, private companies and local populations) and their respective priorities. Indeed, the energy transition is no longer exclusively an abstraction embodied in the targets of official documents (such as the European Green Deal and the PNEC); on the contrary, local populations are becoming exposed to its socioeconomic and environmental impacts in their daily lives. Increasingly aware of the justice and social equity implications of RE transitions, scholars have been progressively engaging with bordering work on justice in energy affairs, contributing to the convergence between EJ and just transition literatures [32–36]. Studies explored how injustices are embodied in RE transitions, potentially reproducing and/or aggravating inequalities, vulnerabilities and exclusions inherited from fossil fuel-based systems [e.g.,37]. By analyzing four examples in Europe, Sovacool et al. [38, p.11] identified 44 injustices transcending geographic scales and life-cycle stages, with Southern countries bearing the brunt of European low-carbon transitions.

Banerjee et al. [39] assessed the extent to which literature on renewable electricity addresses EJ, stressing that benign narratives about RE can disregard the voices of the rural periphery – preferred locations for installing RE projects – as well as the cultural, economic and environmental ties communities have to their lands [39,p. 781]. Studying a RE project in Vietnam, Siciliano et al. [40,p.11] explored how different segments of the Vietnamese society disagree regarding environmental and socioeconomic priorities: while local people affected by the RE project care for its socioenvironmental impacts and procedural justice, institutional actors are more concerned with climate change mitigation, greenhouse gases abatement, energy security and economic growth.

Beyond an analytical tool, energy justice can also be interpreted as the materialization of alternative energy futures built from below. Rasch and Köhne [41] analyzed the opposition to shale gas development and the preference for RE production in the Noordoostpolder, the Netherlands. The authors [41,p.608] claim, on the one hand, that the analysis of EJ should be contextualized, taking history, time and location into account; and, on the other hand, that EJ is constructed from below, stemming from people's RE practices – what they consider the "just" alternative –, which entails new imaginations of EJ. Thus, to ensure "ownership of the energy transition", EJ related policies must include ideas and practices of local actors engaging in energy [41,p.613]. We contend that this "engagement in energy" should also encompass resistance and opposition to RE deployment when EJ principles are not met.

Despite this burgeoning empirical research on EJ, literature concerning the justice implications of solar energy development is still scarce. Exceptions are the study of procedural and distributional injustices associated with solar energy in India [42,43] and community acceptance of large-scale solar farms in Great Britain [44]. Regarding the Alentejo region, where our case study is situated, the study of Campos et al. [45] is worth noticing, although it does not rely on the EJ framework. The authors [45] focused on the degree of citizens' acceptance of solar energy installations in Alentejo and Andalucia (Spain) and their association to different sizes of the installations and different degrees of information and participation. Campos et al. [45] concluded that citizens were more satisfied with small size installations, while also presenting moderate to high satisfaction with large-scale solar projects, although populations already exposed to large solar installations, like in Andalucia, were comparatively less satisfied. Results also showed that "citizens were more interested in participating in projects offering benefits to more vulnerable populations" and that transparency and participation are important factors of citizens' satisfaction levels with solar installations [45,p.11].

Nevertheless, given the expected increase in solar energy in southern European countries, EJ, supported by its three main tenets, enables the critical analysis of energy transitions in different socioeconomic and environmental contexts, uncovering their potential contradictions and shortcomings. In this research, it allows us to assess the justice implications of deploying a controversial SPP in a rural area, more specifically, the distribution of ills and benefits between different actors, the suitability and inclusiveness of decision-making processes and the fairness in the representation of all interested parties.

3. Methodology

3.1. Case study

Our case study focuses on the public controversy around the construction of a large SPP near the village of Cercal, in Alentejo, southern Portugal, where local protests over the infrastructure unleashed a public controversy, initiated in May 2021 and are still ongoing. The SPP – which, as of March 2024, was not installed – is planned to cover around 300 ha (137 ha will be covered by solar panels) and produce up to 596 GW/h, which will be transported in 25,6 km of high-voltage lines until the point of connection with the national electrical grid in Sines [46] [Image 2 and Image 3]. Aquila Clean Energy, part of Aquila Group, is the developer of the project, through the company Cercal Power.

On May 7, 2021, the Portuguese Environment Agency (APA), the developer and local authorities (the parish and the municipality) held a public session to inform residents about the project, which turned into a tense and conflictual interaction and eventually led to the creation of a local protest movement against the installation of the SPP. The media covered the case, and several environmental associations and local tourism entrepreneurs expressed their concerns over this project. Two political parties raised questions about it at the Parliament. According to Portuguese legislation, such RE projects require an environmental impact assessment (EIA), presented by the developer, and a public consultation, conducted by APA. In public consultations, every citizen may voice their opinion, which APA then analyzes and addresses issuing a report and may, accordingly, request changes to the project. In Portugal, the public consultation report is mandatory for several projects with environmental impacts and is the only official mechanism for participation in environmental decisions.

Right after the public session on May 7th, a few residents gathered to support each other's participation in the public consultation, studying the official documents of the project and formulating arguments, substantiations, and alternatives. They then formed the local protest movement, which organized three other public sessions in the village to inform and answer residents' concerns and to promote discussion, bringing guests to talk about the subject. The movement is also active on its Facebook page and in the media. During fieldwork, several residents reported they either had been at, or had heard about these sessions, and some consulted the movement's Facebook page for information and updates on the process.

In May 2021, APA issued a conditional approval of the SPP project. The terms included safeguarding all *Quercus* trees,[1] guaranteeing the free flow of water lines, conserving gallery forests, excluding slope areas and areas where houses face the plant directly, implementing a curtain of trees in all extension of the panels' perimeter, and obtaining additional permissions from state administration bodies [46]. Besides, the developer should present monitoring plans regarding invasive species, erosion control, temperature control, avifauna, and sound environment [46]. In October 2021, the local protest movement initiated a lawsuit to suspend the administrative act of the Environmental Impact Declaration on the project. As of March 2024, the Court had not issued a decision.

On October 19, 2022, Aquila Clean Energy, the Faculty of Sciences of the University of Lisbon and FCiências.ID, a private non-profit organization, signed a two-year cooperation agreement by which a team of researchers from the Faculty of Sciences of the University of Lisbon will monitor the impacts of the construction of the SPP on the surrounding temperature, as well as on soil quality and biodiversity, with the aim of defining strategies for preserving local biodiversity and mitigating potential negative impacts [47]. On January 10, 2023, Aquila Clean Energy signed a partnership with the University of Évora, to develop an agro-photovoltaic pilot project in Cercal's SPP, covering both areas in-between solar panels and surrounding unused areas [48]. Aquila Clean Energy affirms its intention to address the concerns expressed by the population, thus seeking greater local acceptance of the project [48]. In March 2024, researchers from the Faculty of Sciences, University of Lisbon, communicated the first results of their study to the developer [49]. The researchers claim they now have an exhaustive characterization of the area, which will serve as reference to assess impacts of the SPP. They conducted physical, chemical and genetic analysis of the soil, and identified plants, invertebrates namely pollinators, birds, mammals, amphibians and reptiles [49]. Focusing on these results for a sample land of 10 km2 to be occupied with panels, researchers recommended that the developer recovers and maintains water lines, preserves specific areas, and follows management plans to compensate species under greater threat [50]. (Please see Image 4 for a summary of the main events of the controversy).

Cercal is a parish in the municipality of Santiago do Cacém, in Alentejo, a region characterized by its vast plains, low density population and land ownership concentration, historically portrayed as underdeveloped, and its population often stigmatized as "lazy". During the fascist dictatorship (especially between 1928 and 1938, when the dictatorial government wanted to ensure Portugal's self-sufficiency in cereal production), Alentejo underwent an unprecedented agricultural modernization causing massive soil erosion. During the dictatorship, Alentejo harbored a vibrant network of antifascist workers [51,52]. More recently, especially in the southwest region, intensive agriculture based in greenhouses relies on the work of foreign workers living in very precarious conditions [53]. Alentejo is the second region in the country regarding the share of farming land dedicated to organic farming [54]. Alentejo has also attracted western and northern European citizens, some of whom are engaged in agroecological activities in the region.

In the Census of 2021, Alentejo registered 704,533 residents, being among the regions with lower population density in the country, and the region that lost more population in the last decade [55]. The main activities,[2] as of 2022 are: 1) Social assistance to the elderly, with accommodation; 2) Retail sale in supermarkets and hypermarkets; 3) Freight transport by road; 4) Agricultural service activities; and 5) Growing of crops combined with farming of animals (mixed farming) [56].

3.2. Data and analysis

In May 2022, drawing on media coverage of the controversy, we identified a first set of stakeholders and invited them for an interview. In June 2022, the first author, who is an anthropologist, a woman originally from another region in Portugal, undertook a short period of intense fieldwork to understand the socioeconomic, environmental, and geographical context of the controversy. There we started with a spontaneous approach through local commerce and institutions, followed by snowball sampling. Overall, we conducted sixteen interviews with nine residents, two representatives of the local protest movement, three local elected officials, and two representatives of environmental associations (please see Tables 1 and 2).

According to its representatives, the local protest movement emerged from meetings held by residents and concerned citizens from neighboring villages, who were trying to understand the documents and writing comments for the public consultation. Subsequently, they organized informative sessions and several protest actions against the implementation of the SPP. The movement is informal, there is no formal register of members, rather, people gather around it through participating in actions and following its social networks online. Both the environmental non-governmental organizations (NGO's) we interviewed took a position in the public consultation of the project. One of

[1] Portuguese Law (Diário da República n.° 121/2001, Série I-A de 2001-05-25) protects *Quercus suber* and *Quercus ilex*, regulating all interventions on these trees.

[2] Ranked according to number of employees in establishments.

Image 2. Map of the Alentejo region marked with the blue line, and Cercal marked with a blue dot. Authors' adaptation from SNIG National System of Geographical Information of the Portuguese Republic.

Image 3. Map of Cercal, the blue dot, situated in Alentejo region marked with the blue line, in Portugal and in the Iberian Peninsula. Authors' adaptation from SNIG National System of Geographical Information of the Portuguese Republic.

them is an international NGO with a Portuguese delegation and the other is originally from Portugal and has a national scope.

Our sample reflects the research project's time constraints, which prevented fieldwork from being extended, but also the social context of the case study and the population's distrust and hesitation in addressing any issue related to the SPP. Indeed, several people that were contacted in the parish did not wish to concede an interview. Tourism owners who had been vocal against the SPP refused to be interviewed, one of them stating they were tired from the process. One shop owner initially scheduled an interview but gave up; they contacted a friend who also scheduled an interview and gave up. A resident whom a respondent called to give an interview, upon becoming aware of the nature of the

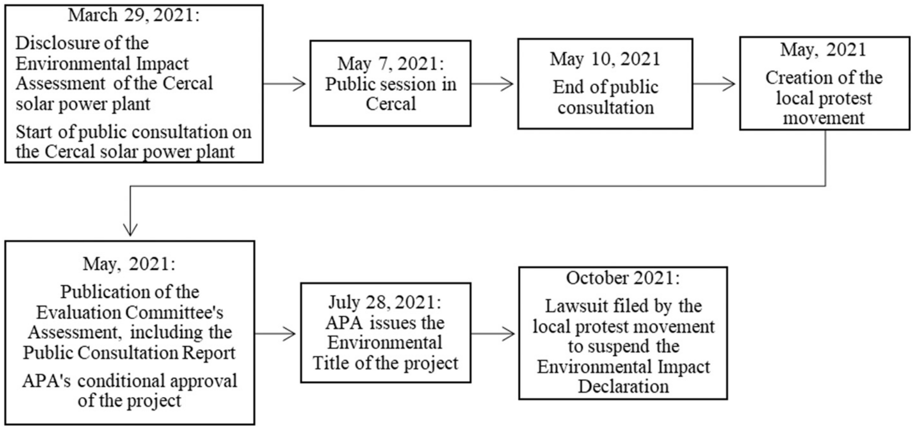

Image 4. Flow chart illustrating some of the main events of the controversy.

Table 1
Actors contacted for interviews.

Respondents
Representative of national environmental association
Representative of international environmental NGO (Portuguese delegation)
Representatives of local protest movement
Parish elected representatives
Municipality elected representative
Nine residents
Actors/ entities who refused to be interviewed
Company developing the project
Portuguese Environment Agency
Regional Territorial Planning Office
Directorate General for Energy and Geology
Fire Department
Ten residents

Table 2
Charaterization of respondents.

Respondent	Age	Gender	Occupation	Place of interview	Interview code
Representative of national environmental association	46	Man	Responsible for 'Policies Follow-up' at the NGO and member of its directive board	Online	EA1
Representative of international environmental NGO (Portuguese delegation)	42	Woman	Responsible for the 'Policies Section' in the NGO	Online	EA2
Representative of local protest movement 1	71	Man	Retired field geologist and environmental advisor for mining companies	Online	LPM1
Representative of local protest movement 2	31	Woman	PhD student	Online	LPM2
Parish elected representative 1	50	Man	Construction work technician, construction oversight and design	Online	PE1
Parish elected representative 2	60	Man	Public school transport driver	Face to face	PE2
Municipality elected representative	46	Man	Lawyer	Online	ME1
Resident 1	Unknown age	Man	Agriculture worker and firewood seller	Face to face	R1
Resident 2	41	Woman	Hairdresser	Face to face	R2
Resident 3	30	Woman	Waitress	Face to face	R3
Resident 4	29	Woman	Waitress	Face to face	R4
Resident 5	19	Man	Student	Face to face	R5
Resident 6	47	Man	Hodman	Face to face	R6
Resident 7	67	Woman	Retired agriculture worker and hotel worker	Face to face	R7
Resident 8	86	Woman	Seamstress	Face to face	R8
Resident 9	74	Woman	Household worker	Face to face	R9

topic walked away upset. Moments later, the interviewer crossed paths with this person (it is a small village) and apologized for any disturbance. The resident indicated that such theme (the SPP) was something they would rather keep to themselves. A wine bar attendant outright refused to give an interview, stating the wish to avoid such a divisive issue at the local level. One of the respondents asked not to be filmed or recorded, because the controversy had brought them enough exposure and discord with neighbors. Three other residents declined the request for an interview, claiming their ignorance on the theme, and the firefighters refused an interview claiming that at that stage it was not up to the fire department to make a statement on the project. Although short, the immersion in the local context allowed us to understand the impact of the controversy on local social relations and on individuals, which explains several people's unwillingness to be interviewed. Despite our repeated contacts, state agencies and the developer did not respond (Please see Table 1).

Given these constraints, the sample is limited and apparently biased against the SPP. However, the local protest movement comprises over one hundred people in a parish with 2954 residents [55], which is not neglectable. Moreover, most interviewed residents, although voicing doubts and/or specific criticisms, did not have a polarized position against the SPP, and some were in favor because it will produce clean energy. Similarly, the parish elected officials expressed criticism on some aspects, but did not stand against the SPP. Therefore, we consider the data is representative of the different actors and nuanced perspectives shaping this controversy.

The semi-structured interviews followed a script encompassing respondents' knowledge of the SPP and its impact in the region, the consultation process and views for the future of the region, with some differences according to types of actors.

All participants provided written or oral consent, and all interviews were recorded, transcribed, and anonymized, apart from one (the respondent only agreed to written notes). We also drew on media analysis from national, regional, and local newspapers, a video recording of the public session posted online and APA's official documents, namely the EIA and the public consultation report.

We conducted a thematic analysis [57] to identify the application, or lack thereof, of the three tenets of EJ – distributional, procedural and recognition –, as proposed by McCauley et al. [21]. More concretely, we carefully read all interview transcriptions and identified the emergent categories from the respondents' answers: 'land', 'sacrifice zone', 'impacts of the SPP', 'information', and 'consultation'. We then assigned these categories to the corresponding tenet(s) of the EJ framework. For instance, 'land' was mentioned several times and this category related to different EJ tenets: the secrecy of the land leasing process relates to both procedural and recognition justice, whereas land uses and landscape changes relate to distributional justice; consequently, we discuss 'land' in light of the three tenets. We proceeded in the same way for the other categories, assessing whether EJ tenets are accomplished or not. As mentioned by Liu [57], when looking at our categories 'against' the thematic categories of a theoretical framework, we need to be attentive to matches as well as to discrepancies, as they are equally revealing. In this case, during the analysis it became clear that recognition and procedural justice are more entangled in practice than their distinction in the literature suggests. We therefore explore this finding in the presentation and discussion of the data.

While we kept record and explicitly identify the actors who voiced the categories, our goal is to identify all the 'salient' categories in the data even if they were not high in frequency [57]. The limited sample also determined that the criteria of category 'frequency' was necessarily fragile. Therefore, more than mapping the more frequent categories or identifying polarized positions in favor or against the SPP, we attempted to trace relevant issues springing up in the field regarding energy justice, indicating points of improvement and interrogation on current practices of energy transition in Portugal.

4. Results

Our analysis follows the three tenets of EJ proposed by McCauley et al. [21]: distributional, procedural and recognition, illustrating how these dimensions unfold in this specific case study. Distributional justice refers to the degree in which the allocation of benefits and harms, and the associated responsibilities of energy deployments, are equally distributed in society. Procedural justice assesses the extent to which engagement and decision-making procedures adequately include all interested and affected, and guarantee transparency. Finally, recognition justice looks at the ways in which people and groups are recognized and included – or not – as having a right to participate in information and decision-making processes. This entails looking at misconceptions, misrepresentations, prejudices and/or exclusion of specific people and/or collectives who have stakes in the matter [21]. (Please see Table 3 for a summary of the results)

4.1. Distributional justice

Regarding distributional justice, three relevant dimensions are salient in this case study: the perceived transformation of Cercal into a "sacrifice zone", especially in the considerations of the local protest movement, the negative environmental and social impacts of the SPP, as mentioned by most of the respondents, and contested land uses.

4.1.1. Sacrifice zone

In February 2023, APA approved a larger SPP (covering more than 1000 ha) in the same municipality of our case study. Desertification is a threat to the rural areas of Alentejo [58], both in the sense of soil and biodiversity loss and population loss. In the local protest movement's perspective, the concentration of solar power plants in the region cumulatively accentuates its social, economic, and environmental vulnerability, turning it into a 'sacrifice zone' connected to the reconversion of the neighboring industrial-energy town of Sines.

> *«That is why we use the term "sacrifice zone", because we think we're being sacrificed for the development of Sines»* (LPM2).

> *«(…) only four jobs are created in this megalomaniac project, (…) hundreds of direct and indirect jobs are lost, related to farming, livestock, forestry, cork, rural tourism, so, our claim is for something more just. There's also another claim (…), that the risks, the non-benefits, are distributed among all (…), we don't want to be a sacrifice zone»* (LPM1).

The local movement and environmental associations consider that APA did not adequately assess the cumulative and interrelated effects of the projects in the region:

> *« (…) maybe that project in that moment doesn't present a risk, but when you place it in perspective it presents cumulative damage and other alternatives could've been considered and they weren't (…) »* (EA2).

> *« (…) when we're talking about 6 %, 10 % of the territory, it's already a very significant area (…), we need to assess if the impacts will be very significant and it seems to me this is not being assessed, because the assessment is one-off (…) honestly, the cumulative assessments that have been conducted don't say anything (…)»* (EA1).

In contrast, the EIA, which cites cumulative impacts twice (regarding soil occupation and use, and landscape impacts[3]), concludes that flora/biodiversity impacts do not accumulate [46].

4.1.2. Negative environmental and social impacts

Residents fear an increase in temperature and its adverse impacts – summer temperatures already reach 45 °C. There is high uncertainty on the effects of the SPP on local temperatures. While the representative of the municipality rejects that possibility, the local protest movement and many participants in the public consultation were concerned about a potential temperature rise. APA's final appreciation of the project states that there is still no scientific evidence corroborating this concern and recommends temperature monitorization and spacing out panels to counteract any irradiation effects on fauna and flora.

[3] Regarding landscape impacts, the report considered that the high voltage line negatively added to an already chaotic and overcharged system, and referred landscape artificialization, aggravated by the absence of enough vegetation around the area to mitigate the visual impact of solar panels. However, the report considers that existing mitigation measures are sufficient.

Table 3
Summary of results.

Distributional justice
Green sacrifice zone
Impacts of the regional concentration of large solar stations and mining projects.
Cumulative impacts not assessed in the environmental impact study.
Connections of Cercal SPP to Sines' reconversion not discussed in the project's official documents.
Negative social and environmental impacts
Landscape change and loss.
Loss of biodiversity, forest, and wildlife.
Fear of an increase in local temperature.
Contested land uses
Land as part of local economy and sociability through farming as opposed to land as commodity, and the monoculture of solar energy.
State/collective definition of where to locate solar stations as opposed to market/private driven decisions.
Procedural justice
Land leasing process
Lack of transparency.
Unequal power relations between land owners and developer.
Public information
Insufficient, inadequate, and untimely public information.
Public consultation
Barriers to participation: untimely information; reports' excessively technical language.
Non-binding participatory process.
Recognition justice
Exclusion of residents from public information and consultation.
Dismissal of protest by APA and elected officials.

Deep landscape changes affect the daily relation of residents with the territory, as well as 'nature tourism', a relevant activity regionally and locally. Indeed, in the beginning of the controversy, tourism operators were strongly against the SPP.

Environmental associations, but also one resident and the local protest movement, underline that wildlife will surely suffer. Among the possible damages is biodiversity loss, particularly plants, trees, and wild animals (especially birds who nest and feed in the area).

> *«The little birds, hares, migratory birds who go out only at night to avoid being eaten, they die because they hit the wires»* (R1).

> *«We had wild animals, forests, and now the spaces increasingly disappear, (…) we know all those consequences! Despite that, we go to vulnerable places and continue to do the same! »* (LPM2).

Projects must avoid protected areas for wildlife and biodiversity, like Natura 2000.[4] Still, environmental associations assert that the project affects important transition areas, as animals are unaware of human demarcations:

> *« (…) there's a classified area* [Natura 2000] *with a watering perimeter inside* [which] *was used, in the last decades, in an indiscriminate way to install farming in greenhouses, that is, there is not a true compatibilization with natural values, so, areas outside Rede Natura are important for priority species (…) »* (EA1).

Similarly, APA's assessment must consider the potential biodiversity and forest loss and must comply with legislation regarding protected trees. The project's approval is conditioned to compensation and mitigation measures [46], but environmental associations warn that the areas involved are too impactful:

> *« (…) the distance between the panels is not that significant, the space between lines is often very small, that means the benefits for biodiversity are not that significant. (…) we're advancing to a type of investment that's "we cut the forest here, but there we have pine trees, eucalyptus, those are degraded areas, so let's replace it, let's put up panels, all's good" (…) at the end of the day, we're reducing our forest area, which delivers important ecosystem services (…) we're destroying our carbon sequestration ability to install photovoltaic panels (…)»* (EA1).

4.1.3. Contested land uses

Interviewees expressed worries about losing good farming land. Despite APA's report stating that most of the leased lands were unproductive, this is not the opinion of residents, the local protest movement and parish elected officials.

> *« (…) lands are causing some controversy here because they were farming lands! »* (PE1).

> *« (…) if there was a national territorial planning to this type of plants, certainly Cercal would be out, because here panels will be installed in farming lands, in fertile lands, which will be out of any farming in the next… I don't know how many years, it's forever»* (PE2).

One resident stated that the SPP will occupy some of the most productive lands in the municipality and it should be installed in unproductive land instead.

> *«I don't know, I think maybe it should be unoccupied lands (…) if they placed the panels on top of houses it would not matter, it would not bother us, or even if we look and see the panels it's not that bad, and, maybe, it would not ruin land that is good for farming, for animals and for other things»* (R2).

Deep changes in land use, specifically reducing farming land availability, directly affect the economic structure of a region historically dedicated to farming.

> *«There's the impact on farming land, sustainability, food sovereignty, jobs (…) »* (LPM1).

> *«I heard farmers who (…) were proposed* [to lease their land] *sent them away! They said: "I don't farm because I want short term rewards,*

[4] Natura 2000 is a coordinated network of protected areas in Europe, including its most valuable and threatened species and habitats. https://ec.europa.eu/environment/nature/natura2000/index_en.htm

economic return, that's not what we want. We take care of the land; we're the guardians of the land" » (LPM2).

This also raises questions about distinct conceptions of the land. According to the local protest movement, the SPP project turns it into a mere commodity:

« (…) land is a commodity (…), it's a purely commercial vision and that, throughout history, triggered all kinds of conflicts. They arrive with the attitude of "there's space here! This space is unused! We'll use it the way that best suits us", on top of all, to satisfy EU's goals of carbon neutrality » (LPM2).

To residents, land is connected to the local economy of farming, which will be directly impacted by the SPP:

« (…) they'll put the panels, use the lands, but what about the people who farmed there? They'll not farm anymore, they'll not make money, they'll not have the straw.[5] *(…) everybody is saying that they have animals, sheep, cows, goats, and they will keep them guarded, they will not leave them in the middle of the panels, because I don't think that works very well. »* (R4).

The price of land leases unleashed a tension over land use and value. One of the residents said the lease is higher than what one makes by farming and livestock breeding. The representative of the municipality considered this an advantage to the recipients, but for some residents it represents a loss. The local movement reported that, except for one landowner who was sorry for leasing, they never managed to talk to landowners:

« (…) I have also tried to talk to the people who leased lands, not that older man that was robbed, but other people and they [the other people of the movement] *told me "people think you're the devil!" because they think they have a solution for easy income and I am taking it away! »* (LPM2).

The local protest movement considers this model of energy production another example of "monoculture for export", with profit concentration for the developer and extensive damage for the territory, and uses the colonization of North America as a metaphor to illustrate this territory's vulnerability to green grabbing:

« (…) Once again, we're producing something in monoculture to export! (…) that's what I'm against the most. It's the reasoning, the monoculture mentality, that linear mentality of arriving like a cowboy, taking everything down, doing our way and the population not saying a thing, we don't care about the 'indigenous' (…); extract everything possible and then, thirty years later, we don't know what will happen! » (LPM2).

Some suggest floating solar as an alternative, keeping the land for farming:

« (…) couldn't solar panels be installed over the dam channels? They would not harm any farming land, (…) and 22,000 m of main channels would be enough for many solar panels and maybe avoid [occupying] *land that may be good to build and for much needed crops»* (PE2).

Both environmental associations and the parish elected officials underlined that developers and markets should not be the ones defining the location of SPPs:

«If, in fact, the government didn't define locations to control a bit more the situation, it should have. Because not having a definition of the territories to implement the [power] *plants, the developers end up looking for landowners where it suits them the most, where they'll have less expenses (…) »* (PE1).

«This is left to the market (…) and developers present the places they want. I also know that it's always the places where land leases are lower, there's no true strategy in terms of renewable energies (…), there should be a strategic EIA at the national level, defining the areas where we could install [RE] *with the minimum impact and not always harm the same people»* (EA1).

They criticize the absence of a national strategy, which reinforces the economic and political inequalities underlying the process and the vulnerability of local populations to land grabbing processes.

In face of the injustices surrounding the project, the protest movement demands alternative pathways for the local energy transition:

« (…) what we see here is the centralizing model, where everything is placed in the hands of a few; it's the same model of oil companies, we're repeating the same story without acknowledging that the true energy transition should take place in our backyard, in our houses, by saving energy. (…) so, there's another claim here, that of energy justice » (LPM1).

«We're now going to (…) demand things, for example, an energy community; we've achieved something unprecedented – a participatory budget, available to citizens – and we'll participate (…), because we have solutions, we don't want to just say no (…) » (LPM2).

This indicates that the local protest movement does not follow a "not in my backyard" approach, rather vouching for an alternative energy transition. Therefore, we witness not only a claim for a just distribution of harms and benefits, but also for paradigmatic changes in political and energy systems.

4.2. Procedural justice

Regarding procedural justice, three relevant processes are underlined: the land leasing process, public information and the public consultation.

4.2.1. The land leasing process

Developers of solar energy projects must obtain a production license from the State and prove they have rights over the corresponding land [59]. Before the project became public, its developer secretly negotiated land lease agreements with landowners. Given this secretiveness (contracts are not public), the issue is subject to speculation. One of the parish elected officials heard that the developer had been paying leases to landowners since 2018, something also mentioned by members of the local protest movement (although we were not able to confirm the veracity of these claims). One of the residents considered the secrecy as an indicator of the developer's bad faith:

«What we heard at the time, the feedback I got from the people at the information session, was that it was all set up during the pandemic, landowners were contacted, and contracts weren't to be disclosed, that shows bad faith on their side, right? » (R2).

The exact value of the leases is unknown. The local movement is concerned that values may differ according to landowners' negotiation power. Specifically, they referred that more vulnerable landowners, especially older people who do not know how to read or write, should have had support in such negotiation:

«(…) the government should've, at least, negotiated the contract. How did they let a private company speak to people who do not know how to read and write? Without acting, without mediating (…) » (LPM2).

[5] Straw production is an important activity in the region.

This also relates to recognition justice – even in stages preceding an official project, equal and fair conditions among landowners should be guaranteed. Additionally, such procedures had visible impacts on local social relations, usually already delicate regarding land issues:

> *«People here interact badly with each other when it comes to land issues (…) I don't know what it is, normally, people get along fine, until land is mentioned, you speak of land and the friendship ends»* (R4).

Indeed, a resident whose house will face the panels, and was vocal against the project, mentioned that a neighbor, who leased land to the developer, is no longer on speaking terms with them.

4.2.2. Public information

APA publicly discloses the project information as part of the public consultation process. For clarity purposes, we will first consider the process of public information, which entailed several controversial aspects: the secrecy surrounding land lease agreements; the late and inadequate disclosure of the project and the public consultation; the highly technical character of the project's information; and the incomplete disclosure and manipulation of information.

The public consultation of the project was first announced by notices posted on the parish's showcase. This, however, did not significantly reached residents; in fact, the first time the population heard of the project was through the announcement, on the parish's Facebook page, of a public session, requested by local authorities, to be held by APA and the developer on the 7th of May 2021. While this was the main vehicle of information, one of the affected residents stated that a member of the parish personally went to their house to inform them of the session.

> *«I think the way the law is made should be enough. But we know (…) it's not. For example, public consultations are announced online, but not everyone has* [internet] *access. (…) there should always be a face-to-face public information session. This is not mandatory, only when it is demanded (…)* (PE1).

> *«People must be informed, that's why we have this policy of always clarifying when we do something that we feel may create discussion (…) »* (ME1).

Nevertheless, according to the local protest movement and a parish elected official (who admitted their responsibility on the matter), the information on the public consultation process excluded several residents, given that people do not usually go to the parish premises, neither to its Facebook page, and a considerable part of the population does not have access to internet or uses it regularly. Another aggravating factor was the timing of the public session, only three days before the closure of the public consultation. The municipality representative explained that it was due to the need to wait for recommendations regarding pandemic restrictions.

Besides its publication through paper notices, APA announces public consultations in a dedicated webpage. This, however, may also exclude those without internet access, namely the elderly population:

> *«A webpage?! Many people in Cercal don't have internet. Don't they count? »* (LPM2).

A resident whose house will face the solar panels believed the municipality should have informed people adequately, because some of their older neighbors are convinced that, if the SPP is not built, they will lose electricity in their homes:

> *«There are ladies who think that if they don't put the panels here, they'll no longer have light in their homes!* [The energy] *is not even for here! »* (R1).

The environmental impact report, elaborated by the developer according to APA's requirements, gathers the official information on the project. The local movement and the environmental associations consider it too long and technical to allow for adequate information to the public. It has over one thousand pages and contains a forty-nine-page non-technical summary [60]. The highly technical language imposed by more powerful parties is a barrier to participation [61,62]. Research on public participation in water management shows that the centrality of technical reports is common to other public consultations in Portugal, defining the content and form of participation, framing the debate and providing the main repertoire for communication [63,p. 150].

Lastly, we underscore the incomplete disclosure and manipulation of information. The local movement claims that the final use of the energy to be produced in the SPP should be disclosed, so it can be publicly discussed. They argue the energy will not meet local needs, rather, it will feed an energy-intensive data center and a hydrogen factory[6] planned for the nearby town of Sines, where the high-voltage line will connect to the national electrical grid. Sines is an important seaside industrial-energy pole where recently the government shut down a coal-fired power plant. The representative of the municipality also refers this in their interview:

> *«It's obvious we're very attractive, because we're near Sines, where they're planning or intend to build big projects that must be fed by RE, namely green hydrogen and the datacenter (…); another project on the table is a green steel plant (…) and, because energy transport bears a significant cost, the closer to the source the better»* (ME1).

Such information is absent from the EIA. Besides incomplete disclosure, the local protest movement accuses the developer of manipulating information to persuade the public of the project's benefits. In July 2022, the developer made a press release stating that APA approved the project and that significant changes were on the way [64]. However, APA had already approved the project in May 2021 and released no further documents. The local protest movement claims that the developer released the news to induce the public to think the project was significantly improved and consequently approved. They noticed a demobilizing effect on residents, who believed the project would indeed go forward. Furthermore, they argue there are no binding official documents against which the developer may be held responsible for the announced improvements.

With the intention to provide alternative and more complete information to the population, the local protest movement organized three public sessions in the village and created a Facebook page with updates on the project, videos of sessions and testimonials.

4.2.3. Public consultation

Citizen participation in public consultation processes in Portugal is not binding; APA may decide if and which commentaries it addresses by changes in the project. Therefore, it faces no constraints in its final decision other than legal criteria. The first public session demonstrated this tension between knowing about the project, expressing opinions, and not being able to shape the decision-making process. In the video of the public session posted online, one APA engineer claimed the project was approved regardless of residents' agreement, leading them to question their presence in the public session. The residents' protests were visible in the video and again stated in some interviews.

> *«We felt the session was very late. Everything was sort of already approved, everything decided (…). Then, to be able to have a counter*

[6] The hydrogen factory underwent the same process of EIA, and a public consultation in the period 2023-04-19 to 2023-06-01, and the datacenter underwent public consultation from 2023 to 05-26 to 2023-07-07.

proposal… all the deadlines were very short, it was almost on purpose, "let's do it late because there's nothing to be done" » (R2).

Environmental associations expose the general shortcomings of the participation processes in RE projects in Portugal:

«*The population is not involved in this process, because the participation space for civil society is only the public consultation on the impact study, in that moment, the project is already defined, the population is not listened to (…)*» (EA2).

«*Now, to present the project in a final stage where the population was completely apart from any discussion and it has one month, three weeks to analyze it, frequently there are complex projects with very complicated language, and people, first, don't have time for it, and, second, often when they know that the consultation ends in half a dozen days, it's no good (…)*» (EA1).

Locally, there is also a questioning of the political process, suggesting that decision-makers should not force the project upon residents who are against it:

«*I think the process is not exactly easy, as we can see in this small case, in a small village. (…) I think everything has a way and they should try to find an easier solution than to force people into something that is against them*» (R5).

«*Well, to start, one cannot take a territory by force like the Directorate-General for Energy and Geology and the Ministry of Environment and Climate Action and State Secretaries, one cannot take it in spite of the population and impose it (…), there needs to be a consensus*» (LPM1).

The local movement blatantly contests the whole process: «*we're not against the renewables per se, but against the modus operandi*» (LPM2). They sought to fill in the gaps of the process by writing to the developer, sending manifestos to local and national political officials (two political parties attended one of the movement's public sessions), and going to court to ask for the suspension of the project. According to one representative of the local protest movement, the Court of Auditors called their movement to a meeting because it wanted to assess to what extent the State is harming itself by, on the one hand, investing in tourism in the region and, on the other hand, allowing SPPs, which will jeopardize that investment by changing the landscape and pushing tourists away.

Finally, it is important to emphasize the articulation of disempowerment and empowerment processes. Disempowerment is visible when residents and the parish elected official state they lacked the knowledge to take a stand on the SPP and express dissatisfaction for not having a voice in the matter. Although most of the interviewed residents considered there was sufficient information on the project, some of the residents who rejected an interview asserted they did not know enough to have a clear position. One interviewee claimed:

«*Everybody says they disagree; I do to, I don't understand any of it, everybody says (…) that it should be installed in uninhabited places, it's what I hear, but I don't understand any of it, it's just what I hear*» (R7).

The parish elected officials also mentioned their lack of knowledge on the subject:

« *(…) we only follow (…), we don't have the competence to analyze and issue a position about it*» (PE1).

« *(…) when they tell me they're going to install a plant, I must trust experts, I can't be deducing that is good or that it makes heat or cold… I can't have that notion if I don't know of things (…)* » (PE2).

Although in his interview the representative of the municipality indicated that he was convinced of the benefits of the project, he voiced the relative ignorance of the municipality staff on the subject, and his trust on the technical views on the matter:

"*(…) we don't have experts on that field. (…) We don't have a university nearby, like in other places in the country, to support us in these situations. We don't. And, therefore, it's up to the government, who has the means to do such a trajectory, which I think they haven't done to this moment (…). The government must do better, namely when projects start to come up in the field*» (ME1).

Nevertheless, in parallel with disempowerment processes, empowerment processes are also noticeable. For instance, they are visible in the residents' self-organization into a social movement, which sought to inform and represent the local community, and in the local authorities' request for a public information session with the residents. Unlike similar projects in Portugal, local authorities asked for a public session during the consultation period, which, even if inadvertently, triggered a political process with much more participation and discussion than usual. The actions of the local protest movement had national repercussions, especially due to the media coverage of the controversy. Indeed, this public consultation registered 184 commentaries and several media published dozens of articles on the subject, deepening and broadening a formerly limited political space regarding the energy transition in Portugal. Confronted with people's concerns and protest, the developer signed agreements with two Portuguese universities, in 2022 and 2023, to study the impacts of the SPP and its possible compatibility with farming activity [47–50].

4.3. Recognition justice

While the literature clearly defines procedural justice and recognition justice separately, our data showed that, in practice, they are intertwined. More concretely, the procedural justice dimension of our data also revealed aspects regarding recognition justice. The processes of informing and consulting residents were exclusionary, revealing the State and the developer's failure to recognize the local community as a legitimate stakeholder. The announcement of the public session was insufficient, untimely, and conducted by inadequate means, which led to a great part of the population knowing of the project and the public consultation when it was about to close. The same happened regarding information on the first public session, resulting in a low attendance. Furthermore, the essential information for discussion in the public consultation – the EIA – is highly technical and dense, therefore inadequate for most of the local population.

Residents' reactions in the video of the public session of May 7, 2021 show their claim for greater recognition as legitimate stakeholders of the project. The session was meant to be merely informative, and some of the residents were enraged to know about the project only when it was already approved. In the interview, the parish elected official acknowledged the shortcomings of the first public session, including the unpreparedness of APA's technicians and the developer to conduct an informative discussion. However, he did not acknowledge the legitimacy of residents' reactions:

«*The parish contacted APA to do a public information session here. (…) Those leading the process were not ready for that kind of reaction, they're not prepared to communicate publicly, they're technicians. And what happened? They couldn't explain anything. There were people who had a less correct behavior*» (PE1).

The municipality representative also failed to recognize the legitimacy of the residents' reactions against the process and the project, adding that they were a minority:

«The people who were very worried with the need for a public session are the contestants, those who went there to cause confusion (…), most of the people weren't very interested in this» (ME1).

In the local protest movement's perspective, residents were not respected, as they were excluded from the decision-making process:

«It was all that thing of going over the population, and I'd like to show the people that they're not nobody (…) so, imagine, in our situation, when there was no involvement of the people, they felt disrespected, delegitimized, and invisible» (LPM2).

Furthermore, they perceive that residents are afraid to voice opposition to the project:

«What I noticed was fear of reprisals (…); people avoided to speak publicly but would come to speak personally. In a village where one works directly or indirectly to the parish or the municipality or has a cousin or a… the declarations [at the public session] *of the parish elected official and the municipality's elected official were very incisive: "do not make noise because that can be harmful" »* (LPM1).

The municipality representative did not recognize the legitimacy of the movement, claiming that it does not express the general opinion and that most of its members are not locals:

«Most people weren't very interested in this. Maybe they're wrong, obviously, because this is related to their community, but the people who are against [the SPP] *are half a dozen»* (ME1).

« People that are from Cercal are probably very few. (…) even because the people contesting don't spend a cent in Cercal, don't live their life in Cercal, don't buy in Cercal (…). They don't do anything special in Cercal, don't belong to the associative movement, don't contribute to the community and now they're against what can be good to Cercal» (ME1).

This perspective on the movement relates to the fact that some of the protesters are foreign citizens who live in Alentejo.

Besides failing to adequately recognize residents as legitimate stakeholders, APA technicians and the elected officials dismissed residents' protests in the public session and subsequent actions of the local protest movement, also failing to recognize protest and active discord as legitimate forms of public participation.

5. Discussion

This case study shows energy injustices – distributional, procedural and recognition – in action. Regarding distributional justice, the costs and benefits of this project are unevenly distributed among local (the community), national (the Portuguese government) and corporate (the developer company) actors. Interviewees, especially those from the local protest movement, see the SPP as reproducing systemic socioeconomic inequalities within Portugal and the EU. They claim their region will be "sacrificed" to serve broader projects of energy transition led by the Portuguese government, private corporations, and the EU, who will thus attain their carbon neutrality goals, while enabling green grabbing and capital accumulation.

Systemic procedural injustices reinforce this divide between EU, governmental and corporate projects on the one hand, and the "local" on the other hand. Public consultation processes foster an incomplete and belated participation of residents, removing them from decision-making, often replicating the deficit model in public understanding of science [65], which posits that local and lay resistance to sociotechnical endeavors stems from ignorance, rather than from sustained political and logical reasoning. Indeed, public participation literature is clear in stressing that information and consultation are limited forms of participation [66].

Therefore, in this case study, procedural injustice feeds the top-down and corporate capture of energy transitions, failing to address local populations' and environmental associations' concerns and expectations to participate in decision-making. The process is not static, however, and the developer of the SPP took steps towards addressing the residents' concerns later, in October 2022, signing an agreement with the Faculty of Sciences of the University of Lisbon, to conduct research on the impact in soil quality and biodiversity, and implement the monitoring of local temperature [47], and in January 2023, establishing a partnership with the University of Évora to conduct a pilot study on the compatibility between farming activity and the production of solar energy in the area destined to the station [48]. In March 2024, the research team from the Faculty of Sciences of the University of Lisbon presented the first results of their study with a soil and biodiversity portrait of the area before any intervention of the SPP, which will serve as a basis to assess its impacts [49,50]. The local protest movement, although recognizing the study as a positive step, emphasizes its limits - for instance, it excludes the areas affected by the high voltage line - and its inability to prevent the irreversible negative impacts of implementing the SPP [50].

Recognition injustice is inevitably entwined with distributional and procedural aspects, as State administration and the private corporation did not recognize local populations and social movements as legitimate actors to deliberate on the project. Furthermore, they did not recognize the legitimacy of protest and discord as forms of political participation and action. Both environmental associations and the local protest movement are clear in criticizing the mechanism of public consultation as ineffective and unable to provide local populations with a decisive voice in the process. Without adequate recognition, human and non-human populations are "sacrificed" under the aegis of the hegemonic energy transition, and decision-making are exclusive prerogatives of State agencies and private companies, which the local population counter through resistance and alternative procedures. Indeed, the local protest movement attempted to compensate the limitations in recognition justice by promoting access to information and by empowering residents for participation.

This case study not only illustrates the failure in delivering energy justice, but also the shortcomings of energy decision-making processes in Portugal. Committed to enforcing a hegemonic "energy transition" – a profit-driven energy expansion model in which energy production and infrastructure ownership remain centralized –, regardless of its environmental and socioeconomic impacts at the local level, private corporate actors and national and European Union political institutions dominate political decision-making, disregarding issues of due process, good governance, responsibility, grassroots resistance, and sustainability [8,11]. The case study emphasizes a clash between the European Union's vision of a "just" energy transition, supported by the Green Deal legislation, and local materializations of the energy transition. Indeed, in this case study, participatory mechanisms are not sufficient to decisively recognize – and integrate – local populations. While this emphasizes the need to institutionalize binding mechanisms of citizen engagement with the energy transition, it also indicates that RE infrastructures put in place to support climate action are creating new tensions with rural communities and with their views and experiences of the land [9,39].

Our case study also demonstrates how the energy transition draws urban and rural areas apart, deepening long-term socioeconomic tensions and inequalities within Portugal. As other socioenvironmental controversies, such as lithium mining in northern Portugal, have shown [8], State administration and corporate actors frequently see the rural as, simultaneously, a target space and a barrier to the energy transition due to local resistance. This controversy has been considered emblematic of "green extractivism", generating strong local resistance and reproducing energy injustices in Northern Portugal [8].

The SPP planned for Cercal exemplifies the contradictions and necropolitics of the hegemonic energy transition, while simultaneously highlighting Alentejo as a long-standing sacrifice zone, historically instrumentalized as a site of intensive agricultural production and, more recently, as a target space for deploying RE projects.[7] In fact, the local protest movement actively mobilizes the concept of “sacrifice zone” [16], indicating its capacity to make sense of multiple injustices and the contradictions embodied by the local energy transition. After the 1974 revolution that ended the fascist dictatorship, Alentejo was the epicenter of the agrarian reform, with peasants occupying large agricultural territories that were in the hands of a few private owners. This right of lawful occupation was inscribed in the 1976 Constitution of the Portuguese Republic, emphasizing how processes of democratization led to a shift in social property relations. This post-revolutionary ideal politicizes land rights to this day and fuels the local resistance against processes of green grabbing unleashed by the development of RE infrastructures. Moreover, from the 1990s and early 2000s onwards, Alentejo has been a breeding ground for communities and movements with a relevant role in socioecological transitions, functioning as a hub for individuals – often from Northern Europe– who find in that territory an opportunity to develop alternative lifestyles and forms of social organization. This includes ecovillages such as Tamera [67], several initiatives linked to the Transition Network, such as “Transição São Luís” [68], and the civic movement “Juntos pelo Sudoeste”, that has recently raised concerns about the negative impacts of intensive agriculture in southwestern Alentejo. These examples reinforce Alentejo's historical and contemporary role in the politicization of land rights, indicating that our case study is embedded in a wider sociopolitical trend of local and regional resistance.

Moreover, interviewees display distinct levels of “resistance” and forms of criticism towards the SPP. The local protest movement emphasizes its socioenvironmental and health impacts, issues of property and land grabbing, while criticizing the underlying model of energy transition. In the same vein, representatives of environmental NGO's, who have a national perspective on the matter, question the dominant narrative of “green transition” and existing participation procedures. Local politicians, for their part, highlight the absence of a national strategy to implement RE projects. Thus, one could argue that our interviews configure a “ladder” or hierarchy of resistance to the project, with the local movement exhibiting strong and unequivocal rejection, whereas local politicians, although supporting the initiative, still argue that certain governance aspects should be improved.

The empirical data also suggest that this SPP can be understood as part of a wider tendency of turning European semi-peripheral regions into green sacrifice zones for the implementation of RE projects [8,9,12]. The case study analyzed in this article presents Alentejo as a flagship territory for the perpetuation of “solar extractivism” [69]. This allows private corporations to accelerate “green grabbing”, benefiting from the hegemonic discourse of “energy transition” [70] and its corresponding legal, bureaucratic and administrative apparatus, under a wider socioeconomic context of neoliberal capitalism that enables the free flow of private capital within the EU. Since “climate action” is mostly posited from a technological and economic perspective – relying on “technofixes” and markets –, it disregards the deep-rooted socioeconomic causes of the climate crisis, specifically, capitalism [71]. These dynamics extend to neighboring countries of the EU, such as Norway, where market forces drive the wind power expansion despite the already covered energetic needs of the country [9]. Such expansion is creating green sacrifice zones that exhibit two facets of coloniality: they erase the Saami landscape concept of ‘*meahcci*’ (place-time-tasks), and sideline the Norwegian cultural attachment to the natural landscape expressed through the concept of ‘*friluftsliv*’ (nature fruition) [9: 192]. Therefore, local communities' experiences and their political perspectives of and engagement with the land – and with the energy transition – are marginalized. These processes of green grabbing serve the ecological modernization of capitalism [72], allowing private corporations to eschew local sovereignty and leading to multiple and reinforcing manifestations of energy injustice.

6. Conclusion

Drawing on a controversy surrounding a SPP planned for the village of Cercal, in Alentejo, southern Portugal, this paper explored the contradictions and injustices (re)produced throughout the Portuguese energy transition, which echo similar experiences in Europe and around the globe. This article introduces an original case study that furthers research on solar EJ, particularly scholarship on resistance to large solar power plants. It may also foster public discussion around RE deployment, stressing the insufficiencies of public participation processes, while attempting to lend visibility to actors who are often neglected, namely local communities.

Regarding the study's limitations, the sample is relatively narrow and perhaps unbalanced. Many residents had reservations about giving an interview, claiming lack of information, but also concerns about approaching such a sensitive and contentious topic. Hence, participants who are against the project and, therefore, more willing to speak openly about it, may be over-represented. Furthermore, the SPP is still in the planning stage, which means that we cannot accurately ascertain the full extent of (in)justices, particularly distributional, that might result from its installation.

On November 10, 2023, a political crisis erupted in Portugal. The prime minister resigned due to an investigation of corruption related to lithium mining concessions and plans for a green hydrogen plant and a data center in Sines (both mentioned by the local protest movement and the municipality elected representative). APA's and other public organisms' handling of these matters is under investigation. This judicial inquiry inevitably further “politicized” the energy transition in Portugal, turning it into an even more contentious issue and justifying some of the concerns of social movements mobilizing against RE infrastructures and mining projects. These events emphasize the timely dimension of our article, revealing the urgency to scrutinize how the Cercal SPP and other RE projects are developed.

Thus, in addition to following up on this case study, future research should include a comprehensive mapping of large RE projects being planned or already implemented in Portugal to evaluate if and what injustices are being (re)produced in the context of the energy transition, enabling the comparison of different regional contexts. It will be equally relevant to compare the volume of RE projects in semi-peripheral and central regions, to understand whether the logic of green sacrifice zones applies at the European scale. On a smaller scale, it will be important to understand what factors make local communities more receptive or resistant to the development of RE projects.

Although only one case study was analyzed, we can already draw some policy recommendations from this qualitative research. In order to mitigate the energy injustices observed in Cercal, as well as to avoid their reproduction in future development of RE projects, policymakers need to reconsider aspects regarding transparent procedures, coordination between central and local governments, participation processes and compensation measures:

1) foster a deep social, political and technical discussion of the format and procedure of environmental impact assessments, as they are currently failing to protect the environment and the people;
2) municipalities and parish councils should participate in decision-making processes in their own right, as representatives of their electors, especially when concerning projects that could affect the socioeconomic fabric of the territory, and not act as spokespeople for the central government or public agencies. In addition, local authorities are privileged agents to mediate the processes of implementing RE projects, by ensuring that the population is informed and involved, thus should have the knowledge and human and material resources to do so;

[7] In May 2023, three large solar plants planned for Alentejo were under public consultation, and at least eight others were under analysis.

3) APA should be obliged to disclose the conclusions of environmental impact assessments in public sessions with local populations and their appointed representatives;
4) public consultations should give place to public deliberative processes and not be a mere formality required by law, where participants' views can be disregarded in decision-making;
5) following the implementation of RE projects, the State and local authorities, as well as citizens and their organizations, should monitor any potential energy injustices, which should be mitigated through compensatory measures, either by stopping harmful activities, restoring ecosystems and biodiversity, or by securing alternative sources of income and employment whenever local economic activities (such as agriculture or tourism) are disrupted.

As RE projects expand in Portugal, it will be crucial to ensure the cooperative participation of local communities in all phases, from planning to implementation. Co-benefits of such participatory processes are the inclusion of situated knowledge in decision-making, the promotion of social cohesion, the creation of jobs and the diversification of economic activities. Therefore, RE projects must be harmonized with social, economic, and environmental contexts and give space for community energy to flourish. Indeed, good examples of community engagement in the energy transition are already emerging in Portugal, namely the renewable energy communities of Telheiras (a neighborhood in Lisbon) and Culatra (an island in the Algarve region). Finally, participation alone does not structurally change the socioeconomic and political models underpinning this energy transition – top-down, centralized, driven by large RE companies, and guided by profit. Thus, the meaningful involvement of local communities in the energy transition needs to be accompanied by a critique of these models, as they are the main drivers of energy injustices.

CRediT authorship contribution statement

Oriana Rainho Brás: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vera Ferreira:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **António Carvalho:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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